



Food Waste Estimation Hackathon

Who are we?



■ Antonio Rueda-Toicen

- Computer Scientist
- AI Engineer and DevRel @ Voxel51
- AI Researcher at the Hasso Plattner Institute
- Instructor at NVIDIA's Deep Learning Institute and Berlin's Data Science Retreat



Who are we?



■ Felix Boelter

- Computer Scientist
- MSc. in Artificial Intelligence, University of Lugano
- AI Engineer @ Hasso Plattner Institute
- Interested in robotics, computer vision, and reinforcement learning



Important: Join the Discord channel to communicate with other participants during the Hackathon. **We will share data and code through it.**

Step 1 - Discord Invite

<https://discord.com/invite/fiftyone-community>

Step 2 - Access Channel

<https://discord.com/channels/1266527359511564372/1345119286041116763>

#practical-computer-vision-workshops

[Download Discord](#)

Overall Agenda

- › Food waste estimation from images
- › FiftyOne as a tool for dataset curation and model analysis
- › Image embeddings and their uses
 - › CLIP and DINOv2
- › Integrating FiftyOne with PyTorch training runs for regression problems
- › Object Detection and Image Segmentation with YOLOE

Measuring wasted food at elderly care homes

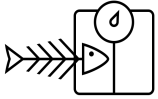
Overhead

1 staff member approx. 15
min/day



**Goal: we want to reduce this
time to seconds**





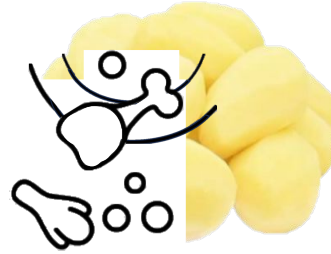
Wasted
Food

Speisenrückläufer: Kartoffeln wiegen

#135411 | Mustermann, Max | SW101 | Schweinebraten , Grüne Bohnen , Butter Kartoffeln , würzige Jus

Bitte füllen Sie die Essensreste des Artikels "Kartoffeln" in den Behälter und übertragen das Gewicht.

**Returned plate: add the ingredient 'potatoes' to the bucket
and write down the weight**



Gewicht des befüllten Behälters:

**Weight of the
filled container**

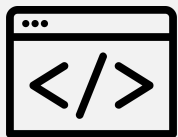


Weigh

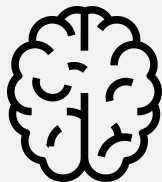


What happens when I
`pip install fiftyone`?

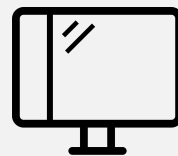
FiftyOne



FiftyOne
Library



FiftyOne
Brain



FiftyOne
App

The FiftyOne Software Development Kit (SDK)

FiftyOne

- > Core library
- > Open source framework
- > Debugger and tracker for computer vision data

```
import fiftyone as fo

dataset = fo.Dataset("my_dataset")

sample = fo.Sample(filepath="/path/to/image.png")
sample.tags.append("train")
sample["custom_field"] = 51

dataset.add_sample(sample)

view =
dataset.match_tags("train").sort_by("custom_field")
.limit(10)

for sample in view:
    print(sample)
```



The core concept in FiftyOne
is the **Dataset**





A FiftyOne dataset connects
all metadata and
annotations to media files

DATASET

Tags

Names

Classes

Description

...

SAMPLE 0

ID

Filepath

Tags

Label

Field

SAMPLE 1

ID

Filepath

Tags

Label

Field

SAMPLE 2

ID

Filepath

Tags

Label

Field

The Dataset Object

 **Dataset:** Think of it as a container for your data, composed of individual Sample objects.

 **Sample:** Represents a single data point (e.g., an image). Stores essential info like filepaths and custom fields.

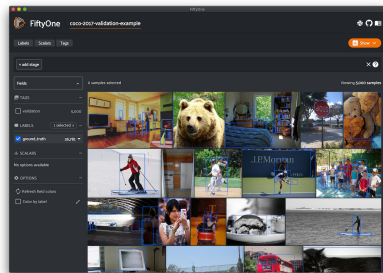
 **Custom Fields:** Add flexibility! Store anything from basic types (numbers, strings) to complex labels (bounding boxes, segmentations).

 **Automatic Media Type Detection:** FiftyOne infers if a sample is an image, video, etc., based on its filepath.

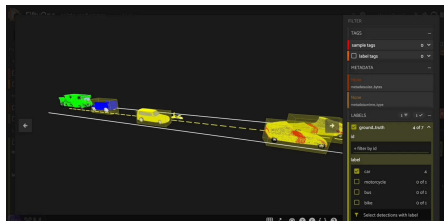
 The Dataset is backed by a MongoDB database



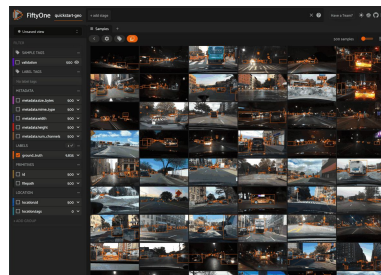
Media Types



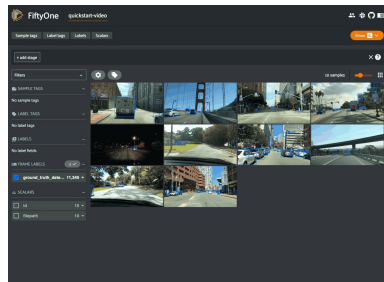
Image



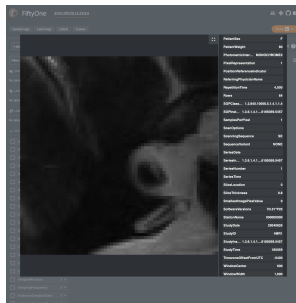
Point cloud



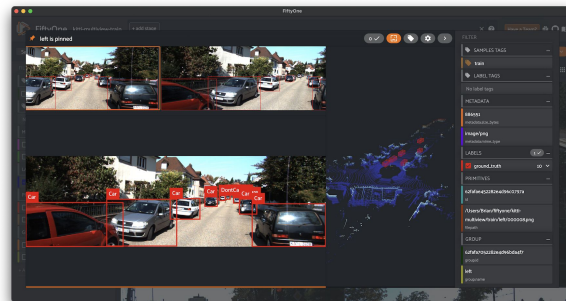
Geo



Video

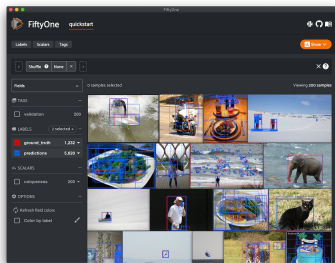


DICOM

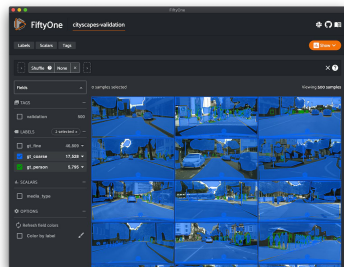


Groups

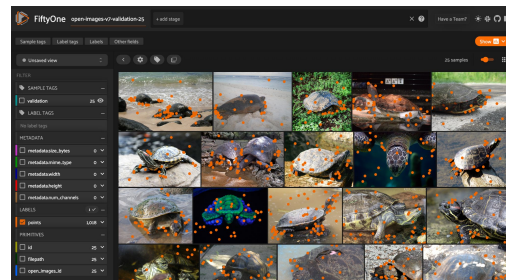
Arbitrary Fields



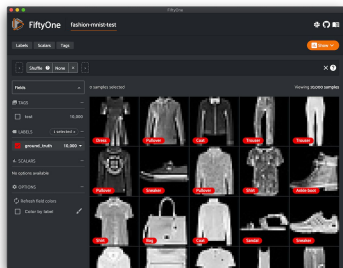
Detection



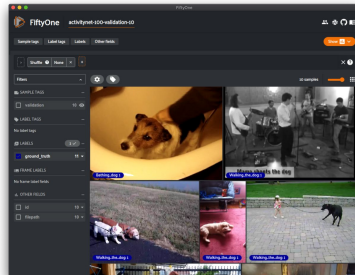
Masks



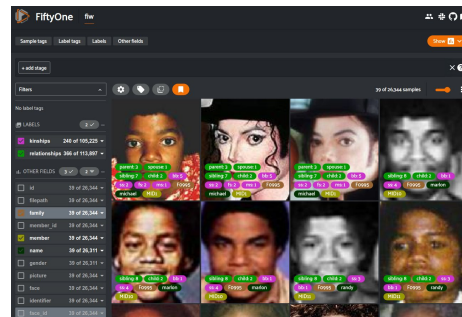
Keypoint



Classification



Action recognition



Relationships and more...



The FiftyOne Brain

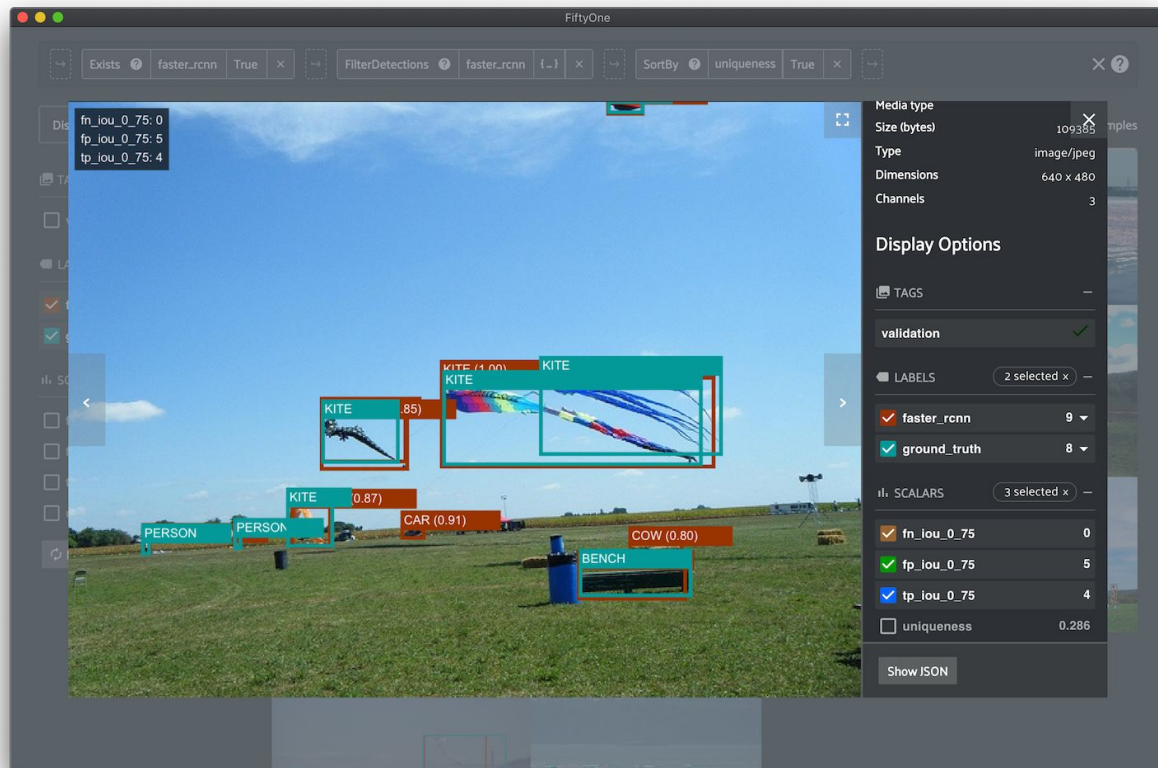
- > ML-powered routines
- > Embeddings
- > Dimensionality reduction
- > Visualization



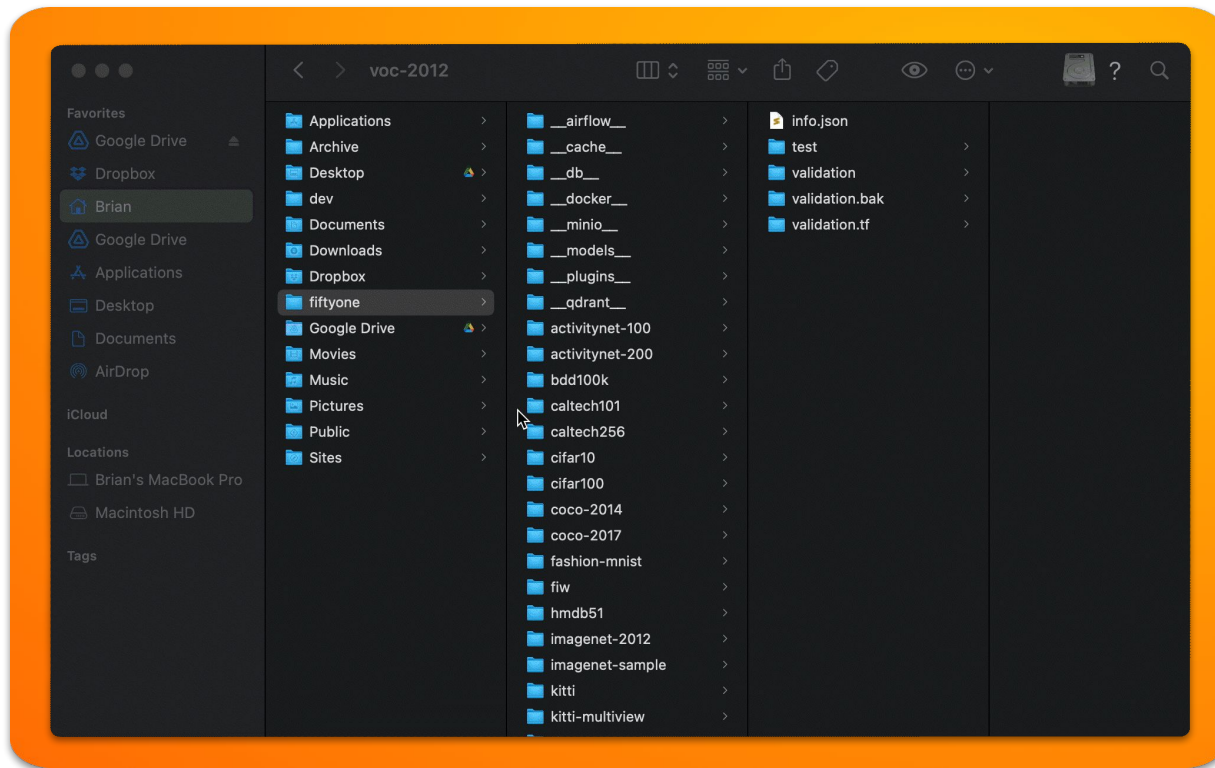
The FiftyOne App

FiftyOne App

- > GUI
- > Application
- > VS Code for your data



Using the App Beats Exploring Data the Old Fashioned Way

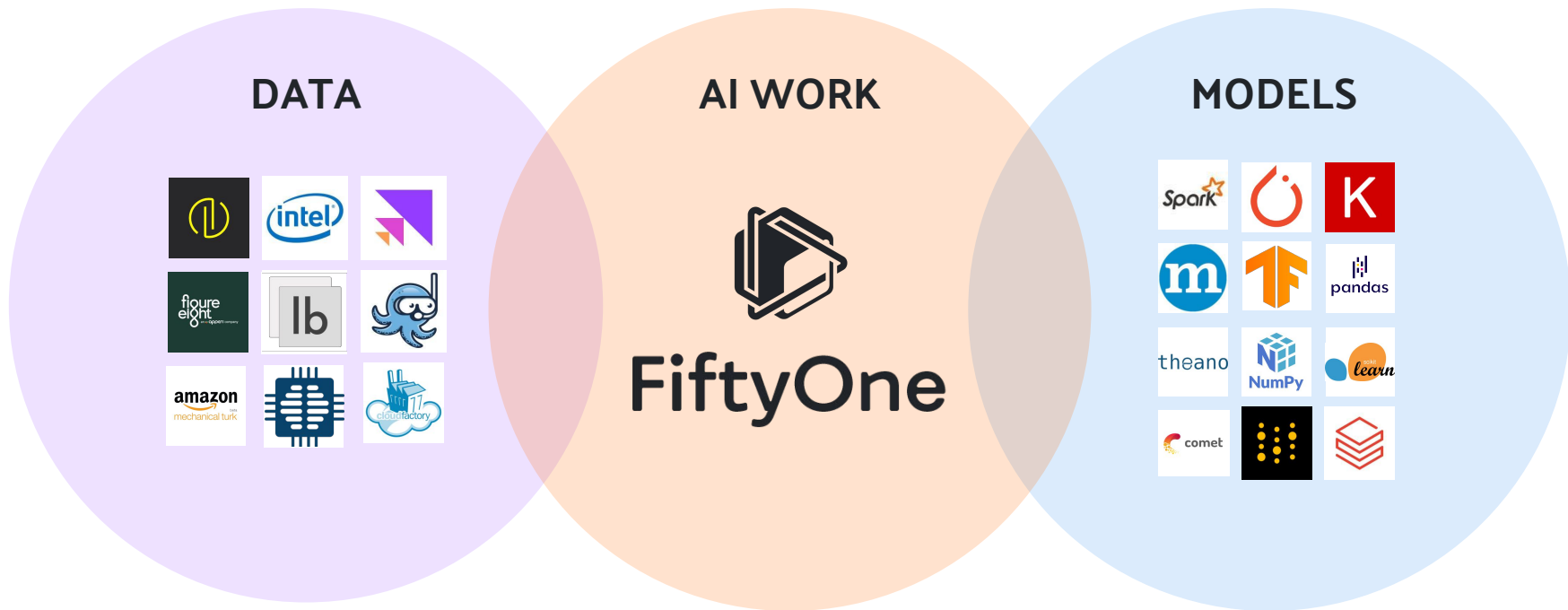


Overview > Core Capabilities > Curate

- > *Find*: filter, match, sort, select
- > *Remove*: duplicates, low quality samples
- > *Add*: tags, metadata, predictions
- > *Correct*: annotation mistakes

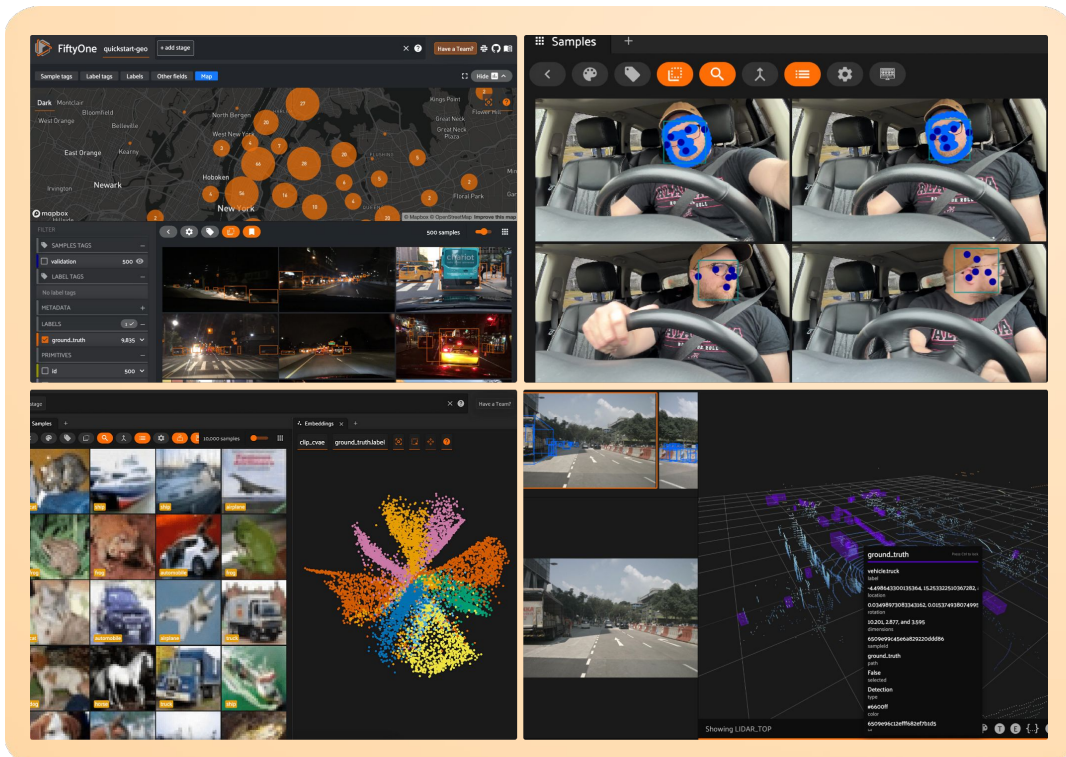


Where FiftyOne fits



Supports all popular vision tasks

- Classification
- Detection
- Instance segmentation
- Semantic segmentation
- Polygons & polylines
- Keypoints
- Point clouds & annotations
- Geolocation
- Multimodal embeddings
- Multiview datasets
- Image, video, & 3D data



FiftyOne Open Source and Enterprise

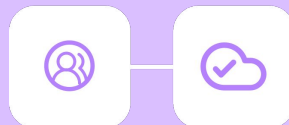
OPEN SOURCE



Enables users to experiment with their projects on a single machine, using local data.

- Visualize & curate datasets
- Find annotation mistakes
- Evaluate & compare models
- Find the root cause of poor-performing metrics

ENTERPRISE



Enables organizations to build & deploy visual AI projects at scale.

All the features in open source, plus:

- Native integration to your cloud or on-prem
- Ability to work with billions of data samples
- Out-of-the-box workflows
- Multi-user team collaboration & access control
- Enterprise-grade scalability & governance

Curating the Food Waste Dataset

1. Adding new samples (40% of the score)

- From new images taken during the event and [shared on our Discord](#)
- From part 2 curated by L. Stroetmann and [published on HuggingFace](#)

2. Adding predictions (40% of the score)

- From our sample notebook
- From methods created by you

3. Feature engineering to improve predictions (refer to our notebook and the FiftyOne documentation) (20% of the score)

- Creating embeddings
 - DINOv2
 - CLIP (we encourage you to do this on your own)
- Adding segmentation masks
 - YOLOE
 - [SAM](#)

Outcome: your own food waste FiftyOne dataset published on HuggingFace



To submit at end of day (19:30 CEST)

1. **Your own food waste FiftyOne dataset published on HuggingFace with all ground truth labels, feature engineering, and predictions from your models (1 per team)**
2. **The Github repository with the notebooks that you used to produce the dataset, its features, and predictions (1 per team)**
3. **The slides (max 3) for your 5 min presentation (add them to Github too as a pdf)**



Useful Notebooks

1. Loading part 1 of the curated dataset ([Notebook link](#))
2. Creating the curated dataset ([Notebook link](#))
3. Creating predictions ([Notebook link](#))



Important features for prediction

1. The image itself
2. `ingredient_name` list
3. `return_quantity` (the amount of grams left on the plate by ingredient)
 - Remember that every ingredient will have a `return_quantity` value

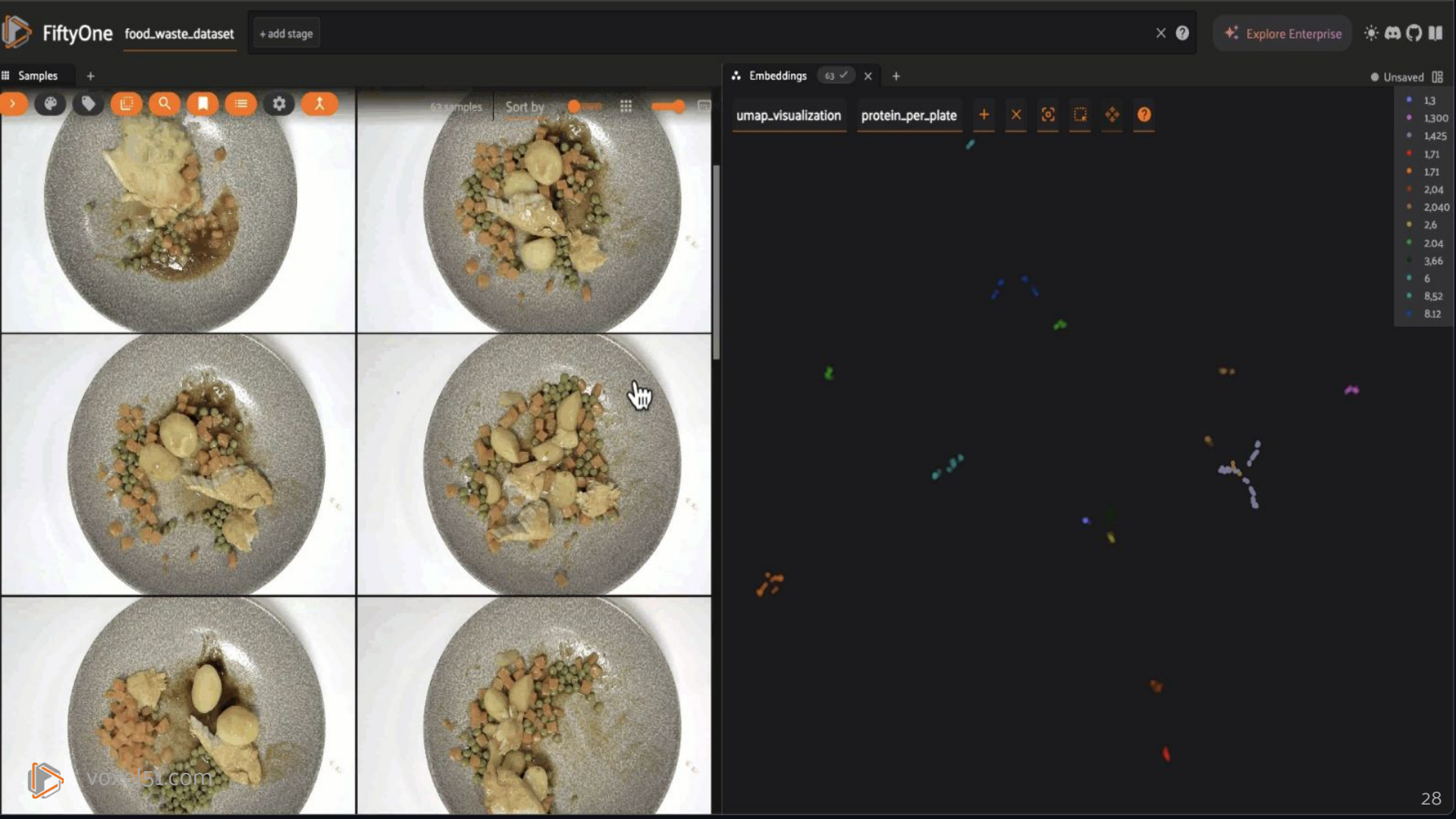


ingredient_name	return_quantity
mashed-potatoes	0
green-beans	26
gravy	14
chicken	40

Getting Help During the Hackathon

1. Ask on the Discord channel
2. Ask the GPT Agent on <https://deepwiki.com/voxel51/fiftyone> (or your favorite LLM)
 - a. We recommend you to use the Gemini LLM on Google Colab





Quick Intro to Dense Vector Embeddings

$$\mathbf{x}_1 = \text{[Image of handwritten digit 3]} \quad \mathbf{x}_2 = \text{[Image of handwritten digit 3]}$$

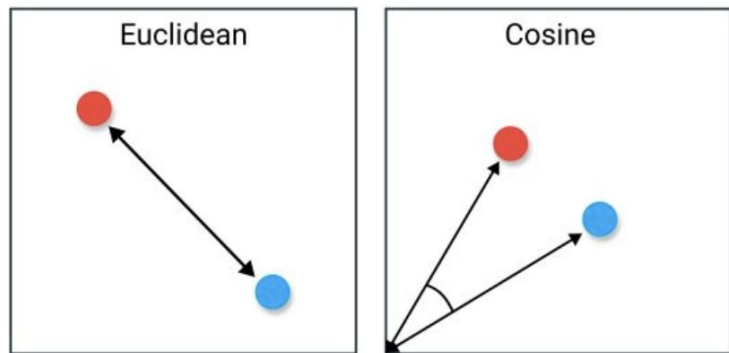
$$\mathbf{x}_1\text{sparse} = \mathbf{x}_2\text{sparse} = [0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^\top$$

$$\mathbf{x}_2\text{dense} = [0.10 \quad -0.75 \quad 0.35 \quad 0.41 \quad -0.20 \quad 0.89 \quad -0.60 \quad 0.03 \quad 0.59 \quad -0.50]^\top$$

$$\mathbf{x}_1\text{dense} = [0.12 \quad -0.85 \quad 0.37 \quad 0.44 \quad -0.22 \quad 0.90 \quad -0.63 \quad 0.01 \quad 0.58 \quad -0.49]^\top$$



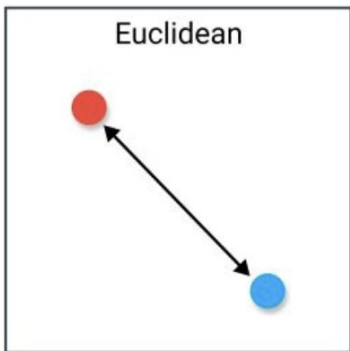
Metrics to Compare Embeddings



$$d_{\text{euclid}}(x_1, x_2) = \sqrt{\sum_{i=1}^n (x_{1i} - x_{2i})^2}$$

$$\text{cosine similarity}(x_1, x_2) = \frac{x_1 \cdot x_2}{\|x_1\| \|x_2\|}$$

Embeddings Are the True Magic of Neural Networks 🧙



CLIP: Contrastive Language Image Pretraining

Learning Transferable Visual Models From Natural Language Supervision

Alec Radford^{*1} Jong Wook Kim^{*1} Chris Hallacy¹ Aditya Ramesh¹ Gabriel Goh¹ Sandhini Agarwal¹
Girish Sastry¹ Amanda Askell¹ Pamela Mishkin¹ Jack Clark¹ Gretchen Krueger¹ Ilya Sutskever¹

- Connects text and images in a shared embedding space
- Created by OpenAI in 2021
- Trained on 400M image-textual description pairs scraped from the internet
- Predicts the most relevant text snippet for an image, enabling “zero-shot classification”

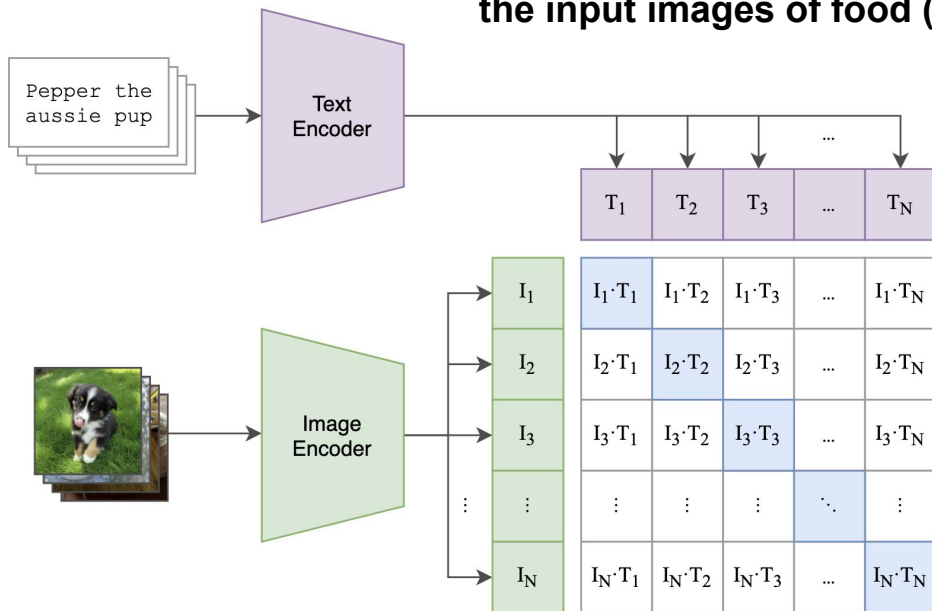
[Paper on ArXiv](#)



How CLIP Embeds Text and Image Pairs

(1) Contrastive pre-training

We will use CLIP to encode both the text from our ingredients list and the input images of food (input features for the regression model)



In many use cases the model is able to create "good enough" labels without extra training

What's next?



[pip install fiftyone](#)



[Book a Demo](#)



[Join our community](#)



[Attend our events](#)



